

Redis is an in-memory data structure store often used as a very fast cache, message broker, and NoSQL database in modern systems.[redis+1](#)

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## 1. What Redis is & key features

Redis (REmote DIctionary Server) is an open-source, in-memory key-value and data-structure store that can act as a database, cache, message broker, and streaming engine.[ibm+1](#)

Key features:[peerbits+1](#)

- Very low-latency reads/writes because data lives in memory.
  - Rich data structures: strings, lists, sets, hashes, sorted sets, streams, etc.[redis](#)
  - Built-in replication, high availability (Sentinel), and clustering for sharding.[severalnines+1](#)
  - Configurable persistence (RDB, AOF, or both) plus optional no-persistence mode.[redis](#)
  - Pub/Sub messaging, Lua scripting, transactions, and atomic operations on data types.[redis+1](#)
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## 2. How Redis handles persistence

Redis is memory-first but can persist data to disk using several options.[dev+1](#)

- RDB snapshots: Periodic point-in-time snapshots of the dataset written to a compact binary file; configured by rules like “after N writes in M seconds.”[redis](#)
  - AOF (Append Only File): Logs every write operation to an append-only log, then replays it on restart to reconstruct the dataset; can fsync on every write, every second, or never.[dev+1](#)
  - RDB + AOF together: Use RDB for faster restarts and AOF for better durability.[redis](#)
  - No persistence: Useful when using Redis purely as a cache and data can be recomputed from another system of record.[redis](#)
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## 3. Redis vs relational databases

Redis is a key-value / data-structure store; relational databases are table-based with SQL and strong relationships.[navicat+1](#)

Key differences:[navicat+1](#)

- Data model: Redis stores keys mapped to simple or complex in-memory structures; RDBMS store rows in tables with schemas, joins, and constraints.
  - Querying: Redis primarily does key-based operations and simple pattern scans; RDBMS support SQL with joins, aggregations, and rich filtering.
  - Performance: Redis is optimized for in-memory, low-latency operations; RDBMS are disk-based (with caching) and optimized for durability and complex queries.[ibm+1](#)
  - Use cases: Redis for caching, sessions, real-time analytics, queues; RDBMS for transactional business data, reporting, relational integrity.[einfochips+1](#)
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## 4. Connecting to Redis (example code)

Most languages have official or popular Redis clients.[dev+1](#)

Example in Node.js using [node-redis](#) (v4):[dev](#)

js

```
const { createClient } = require('redis');

const client = createClient({
  url: 'redis://localhost:6379' // or your Redis URL
});

client.on('error', err => console.error('Redis Client Error', err));

(async () => {
  await client.connect();
  await client.set('foo', 'bar');
  const value = await client.get('foo');
  console.log(value); // bar
})();
```

Basic steps in any language:[geeksforgeeks+1](#)

- Install Redis client library for that language.
  - Create a client with host, port, and optional password/DB.
  - Connect (or let the client connect lazily).
  - Use commands like [SET](#), [GET](#), etc., via library methods.
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## 5. Common use cases in modern apps

Redis is widely used to offload primary databases and handle real-time workloads.[algomaster+2](#)

Typical use cases:

- Caching: Database query results, HTTP responses, and computed views to reduce latency and DB load.[einfochips+1](#)
- Session storage: Storing user sessions for web apps, especially with horizontally scaled servers.[ibm](#)
- Real-time analytics: Counting events, tracking page views, and aggregating metrics with counters and sorted sets.[redis+1](#)
- Message queues / Pub-Sub: Simple queues with lists or streams; Pub/Sub channels for notifications.[ibm](#)

- Rate limiting & leaderboards: Sliding window rate limiters and gaming leaderboards using sorted sets.[redis+1](#)
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## 6. Redis data types with examples

Redis offers several core types, each optimized for specific patterns.[redis+1](#)

- Strings: Binary-safe values up to 512 MB; used for simple values, JSON blobs, counters with `INCR`.[redis](#)
    - Example: `SET user:1:name "Alice", INCR page:views`.
  - Lists: Ordered sequences of strings implemented as linked lists; support push/pop from both ends.[redis](#)
    - Example: `LPUSH log:events "event1", BRPOP log:events`.
  - Sets: Unordered collections of unique strings; good for membership, tags, intersections.[redis](#)
    - Example: `SADD user:1:roles "admin", SISMEMBER user:1:roles "admin"`.
  - Hashes: Field-value maps, like a lightweight object; ideal for storing object attributes.[redis](#)
    - Example: `HSET user:1 name "Alice" age "30", HGETALL user:1`.
  - Sorted sets (ZSETs): Unique members ordered by a numeric score; great for leaderboards, rankings.[redis](#)
    - Example: `ZADD leaderboard 100 "user1", ZREVRANGE leaderboard 0 9 WITHSCORES`.
  - Other notable types include streams (log-like data), bitmaps, hyperloglogs, and geospatial indexes.[redis](#)
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## 7. Implementing caching in a web app with Redis

General cache-aside pattern (language-agnostic):[geeksforgeeks+2](#)

1. Client requests data (e.g., `/user/123`).
2. Application first checks Redis with `GET user:123`.
3. If key exists (cache hit), return the value.
4. If missing (cache miss):
  - Fetch from primary database.
  - Store in Redis with `SET` and TTL (e.g., `SETEX user:123 300 <json>`).
  - Return data to client.

Example (Node.js + Express style pseudocode):[algomaster+1](#)

js

```
app.get('/user/:id', async (req, res) => {
  const key = `user:${req.params.id}`;

  let cached = await client.get(key);
  if (cached) {
    return res.json(JSON.parse(cached));
  }
```

```

}

const user = await db.query('SELECT * FROM users WHERE id = ?',
[req.params.id]);
if (!user) return res.status(404).end();

await client.setEx(key, 300, JSON.stringify(user)); // 5 min TTL
res.json(user);
});

```

Common additions:[geeksforgeeks+1](#)

- Use TTLs to avoid stale data.
  - On writes/updates, invalidate or update relevant cache keys.
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## 8. Benefits of Redis vs other caching solutions

Compared with many traditional cache layers or in-process caches, Redis provides several advantages.[peerbits+2](#)

- Networked, shared cache: Multiple app instances (and languages) share the same centralized cache, unlike local in-process caches.
  - Rich data structures: Native lists, sets, hashes, and sorted sets enable patterns that are complex with plain key-value caches.[redis](#)
  - High availability & scaling: Built-in replication, Sentinel (for HA), and Cluster (for sharding and horizontal scale).[wikipedia+1](#)
  - Operational maturity: Robust tooling, observability, and managed cloud offerings.
  - Durability options: Persistence via RDB/AOF, which many pure caches do not provide.[redis](#)
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## 9. Replication and failover in Redis

Redis uses master-replica replication plus optional Sentinel and Cluster for availability and scaling.[wikipedia+1](#)

- Replication:
  - A primary (master) node handles writes; one or more replicas asynchronously replicate data via a replication stream.[wikipedia](#)
  - Replicas can serve read-only traffic, improving read scalability and redundancy.[wikipedia](#)
- Sentinel (HA / failover):
  - Sentinel processes monitor masters and replicas, detect failures, and automatically promote a replica to master when needed.[severalnines](#)
  - Clients can be Sentinel-aware to discover the current master.
- Redis Cluster:

- Shards data across multiple masters, each with replicas, and handles automatic rebalancing and failover at shard level.[severalnines](#)
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## 10. Common Redis commands and uses

Some frequently used commands:[gist.github+2](#)

- Key operations
  - **SET** key value: Set string value for key.
  - **GET** key: Get string value for key.
  - **DEL** key [key ...]: Delete one or more keys.[geeksforgeeks](#)
  - **EXPIRE** key seconds: Set TTL on a key.
  - **EXISTS** key: Check if key exists.[gist.github](#)
- Hash operations
  - **HSET** key field value: Set field in a hash.
  - **HGET** key field: Get value of a hash field.
  - **HGETALL** key: Get all fields and values.
- List operations
  - **LPUSH** key value [value ...]: Push elements to head.
  - **R PUSH** key value [value ...]: Push to tail.
  - **LPOP** / **RPOP**: Pop from head/tail.
- Set operations
  - **SADD** key member [member ...]: Add members to a set.
  - **SISMEMBER** key member: Test membership.
  - **SMEMBERS** key: Get all members.
- Sorted set operations
  - **ZADD** key score member [score member ...]: Add/set member with a score.
  - **ZRANGE** key start stop [WITHSCORES]: Get members in score order.[redis](#)
  - **ZREVRANGE** key start stop [WITHSCORES]: Reverse order.

## 1. When would you introduce Redis in a system design?

Use Redis when the system is bottlenecked by repeated reads/writes to the primary database and needs very low latency or high throughput. Typical triggers are read-heavy workloads, rate limiting, leaderboards, session storage, and real-time counters or metrics.[hellointerview+2](#)

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## 2. Explain the cache-aside pattern using Redis.

With cache-aside, the application is responsible for reading/writing both the cache and the database. On a read miss, the app fetches from the DB, populates Redis with a TTL, and returns the data; on writes, it updates the DB and invalidates or refreshes cache entries.[linkedin+2](#)

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### 3. Compare cache-aside vs write-through vs write-back with Redis.

- Cache-aside: Only load into Redis on read miss; simple, but first miss is slow and you must handle invalidation carefully.[aws.amazon+1](#)
  - Write-through: Every write goes to DB and Redis; strong consistency but higher write latency.[linkedin+1](#)
  - Write-back: Write only to Redis and flush to DB asynchronously; very fast but risk of data loss if Redis fails.[aws.amazon+1](#)
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### 4. How does Redis achieve high performance?

Redis keeps data in memory and uses efficient data structures plus a single-threaded event loop to avoid lock contention. Commands are simple and mostly O(1) operations, and optional pipelining reduces network round-trips for batches.[github+2](#)

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### 5. What are Redis eviction policies and when do you care?

When Redis reaches its max memory, it can evict keys according to policies like `noeviction`, `allkeys-lru`, `volatile-lru`, `allkeys-random`, etc. You care in cache scenarios to control what gets evicted (e.g., using `allkeys-lru` for a pure cache) and to avoid surprising OOM errors in production.[algomaster+1](#)

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### 6. Describe Redis persistence options and trade-offs.

Redis supports RDB snapshots and AOF logs, which can be combined. RDB gives smaller, faster-to-load snapshots but may lose recent writes; AOF gives better durability but larger files and higher write overhead.[hellointerview+1](#)

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### 7. How does Redis replication work and how does it help scaling?

Replication uses a primary (master) that accepts writes and replicas (slaves) that asynchronously copy the dataset and replication stream. Replicas can serve read traffic to offload the primary and provide redundancy for failover.[vervecopilot+1](#)

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### 8. What is Redis Sentinel and when would you use it?

Redis Sentinel monitors masters and replicas, detects failures, and performs automatic failover by promoting a replica to master. It also acts as a discovery service so clients can find the current master; use it when you need high availability without going full Redis Cluster.[vervecopilot+1](#)

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## 9. What is Redis Cluster? Why not just one big node?

Redis Cluster shards data across multiple masters using hash slots, each master optionally having replicas for HA. This is needed when a single node cannot handle the dataset or throughput; it also improves fault tolerance since some shards can remain available during failures.[leetcode+2](#)

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## 10. How would you design session management with Redis?

Store each session as a key like `session:<token>` with a TTL corresponding to session lifetime. Application servers read/write session data from Redis, enabling stateless app instances and easy horizontal scaling behind a load balancer.[hellointerview+2](#)

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## 11. Design a rate limiter using Redis.

Use a fixed-window or sliding-window algorithm with Redis operations. For example, a sliding window can use a sorted set per user (`rate:<userId>`) with timestamps as scores, trimming old entries and counting current ones atomically to decide if the request is allowed.[getsdeready+2](#)

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## 12. How would you implement a leaderboard?

Use a sorted set where the member is the user ID and the score is the rank metric (e.g., points). Commands like `zINCRBY` to update scores and `zREVRANGE leaderboard 0 9 WITHSCORES` to fetch the top 10 provide an efficient leaderboard.[redis+1](#)

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## 13. What are some pitfalls of using Redis as a primary database?

Redis is memory-based, so costs scale with RAM and data size, and naive designs can exhaust memory quickly. It lacks rich querying, joins, and strong ACID semantics like relational databases, making it unsuitable as the sole system of record for complex business data.[prachub+2](#)

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## 14. How do you handle cache stampede (thundering herd) with Redis?

Mitigations include adding jitter to TTLs, using locks (e.g., `SETNX`-based) so only one worker recomputes a hot key, or using stale-while-revalidate semantics. You can also pre-warm caches and use multi-layer caching (local + Redis) to reduce pressure.[hellointerview+2](#)

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## 15. How would you monitor and debug Redis performance?

Track metrics like used memory, hit/miss ratio, command latency, ops per second, and replication lag using tools like `INFO` and external monitoring (Prometheus, etc.). Use slowlog (`SLOWLOG GET`), keyspace stats, and latency monitors to identify heavy commands, hot keys, or blocking operations.[labex+1](#)

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## 16. How does pipelining help in Redis?

Pipelining batches multiple commands into a single TCP round trip, reducing network latency overhead. It improves throughput, especially for many small commands, but responses are received later in bulk, which can complicate error handling.[github+2](#)

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## 17. How do you design Redis keys in a large system?

Use a consistent, structured naming convention like `domain:entity:id:field` (e.g., `user:123:profile`). Include versioning and sharding hints when needed, and avoid unbounded key cardinality or extremely long keys.[algomaster+2](#)

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## 18. When is Redis a bad choice?

It is a poor fit when data must be durable at all times, the dataset is much larger than memory, or queries require complex relational operations. In such cases, use Redis as an auxiliary component (cache, queue) rather than the system of record.[geeksforgeeks+3](#)

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## 19. How would you scale Redis reads and writes separately?

Scale reads using replicas and read-from-replica patterns; ensure your use case tolerates eventual consistency. Scale writes via Redis Cluster (sharding) and careful key design so load is spread evenly across masters.[vervecopilot+3](#)

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## 20. In a design that already uses Redis, what follow-up questions would you ask?

Clarify the data model, access patterns, required consistency, and durability needs before picking data types and persistence. Ask about expected QPS, size of hot data, acceptable staleness, failover RTO/RPO, and operational constraints (ops team, monitoring, backups).